

Daniel Rho

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Research Interests

computer vision, neural rendering, 3D / 4D representation learning, and inverse physics

Education

University of North Carolina at Chapel Hill

Ph.D. in Computer Science

North Carolina, United States

Aug. 2024 - Present

- Advisor: [Roni Sengupta](#)

Sungkyunkwan University

MSE in Artificial Intelligence

Seoul, Korea

Sep. 2020 - Aug. 2022

- Thesis: "Neural Residual Flow Fields for Efficient Video Representations" (Advisor: [Jong Hwan Ko](#), Co-advisor: [Eunbyung Park](#))

Sungkyunkwan University

Bachelor of Economics & BSE in Computer Science and Engineering

Seoul, Korea

Mar. 2014 - Aug. 2020

- Major GPA (Computer Science and Engineering): 4.44 / 4.5 (top 3%)
- Dean's List, 2018

Publications

CONFERENCE PUBLICATIONS

ProJo4D: Progressive Joint Optimization for Sparse-View Inverse Physics Estimation

Daniel Rho, Jun Myeong Choi, Biswadip Dey, Roni Sengupta

TMLR 2026 - Transactions on Machine Learning Research

NFL-BA: Improving Endoscopic SLAM with Near-Field Light Bundle Adjustment

Andrea Dunn Beltran*, **Daniel Rho***, Marc Niethammer, Roni Sengupta

NeurIPS 2025 - Advances in Neural Information Processing Systems

F-3DGS: Factorized Coordinates and Representations for 3D Gaussian Splatting

Xiangyu Sun, Joo Chan Lee, **Daniel Rho**, Jong Hwan Ko, Usman Ali, Eunbyung Park

ACM MM 2024 - Proceedings of the 32nd ACM International Conference on Multimedia

Compact 3D Gaussian Representation for Radiance Field

Joo Chan Lee, **Daniel Rho**, Xiangyu Sun, Jong Hwan Ko, Eunbyung Park

CVPR 2024 (highlight) - Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition

Coordinate-Aware Modulation for Neural Fields

Joo Chan Lee, **Daniel Rho**, Seungtae Nam, Jong Hwan Ko, Eunbyung Park

ICLR 2024 (spotlight) - International Conference on Learning Representations

Mip-Grid: Anti-aliased Grid Representations for Neural Radiance Fields

Seungtae Nam, **Daniel Rho**, Jong Hwan Ko, Eunbyung Park

NeurIPS 2023 - Advances in Neural Information Processing Systems

FFNeRV: Flow-Guided Frame-Wise Neural Representations for Videos

Joo Chan Lee, **Daniel Rho**, Jong Hwan Ko, Eunbyung Park

ACM MM 2023 - Proceedings of the 31th ACM International Conference on Multimedia

Masked Wavelet Representation for Compact Neural Radiance Fields

Daniel Rho*, Byeonghyeon Lee*, Seungtae Nam, Joo Chan Lee, Jong Hwan Ko, Eunbyung Park

CVPR 2023 - Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition

Regression to Classification: Waveform Encoding for Neural Field-Based Audio Signal Representation

TaeSoo Kim*, **Daniel Rho***, Gahui Lee, JaeHan Park, Jong Hwan Ko

ICASSP 2023 - IEEE International Conference on Acoustics, Speech and Signal Processing

Neural Residual Flow Fields for Efficient Video Representations

Daniel Rho, Junwoo Cho, Jong Hwan Ko, Eunbyung Park

ACCV 2022 - Proceedings of the Asian Conference on Computer Vision

Streamable Neural Fields

Junwoo Cho*, Seungtae Nam*, **Daniel Rho**, Jong Hwan Ko, Eunbyung Park

ECCV 2022 - Proceedings of the European Conference on Computer Vision

NAS-VAD: Neural Architecture Search for Voice Activity Detection

Daniel Rho, Jinhyeok Park, Jong Hwan Ko
Interspeech 2022 - Proceedings of Interspeech

PREPRINTS

MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos

Daniel Rho, Jun Myeong Choi, Matthew Thornton, Biswadip Dey, Roni Sengupta
arXiv preprint arXiv:2605.30320 (2026)

Stop Holding Your Breath: CT-Informed Gaussian Splatting for Dynamic Bronchoscopy

Andrea Dunn Beltran, Daniel Rho, Aarav Mehta, Xinqi Xiong, Raúl San José Estépar, Ron Alterovitz, Marc Niethammer, Roni Sengupta
arXiv preprint arXiv:2604.28179 (2026)

Understanding Contrastive Learning Through the Lens of Margins

Daniel Rho, TaeSoo Kim, Sooill Park, Jaehyun Park, JaeHan Park
arXiv preprint arXiv:2306.11526 (2023)

Professional Experience

Graduate Research Assistant

University of North Carolina at Chapel Hill

Chapel Hill, NC, USA

Aug. 2024 - Present

Research Intern

Lenovo Research

NC, USA

May. 2025 - July. 2025

Research Engineer

AI Tech Lab, KT

Seoul, Korea

Jul. 2022 - Jun. 2024

Graduate Research Assistant

IRIS LAB, Sungkyunkwan University

Seoul, Korea

Sep. 2020 - Jul. 2022

Undergraduate Research Assistant

IRIS LAB, Sungkyunkwan University

Seoul, Korea

Jun. 2019 - Aug. 2020

Research Projects

Next-gen 3D Modeling of Endoscopy Videos

National Institutes of Health, USA

USA

2024-2026

Deep Learning Techniques for Multi-Intelligence using Drones

Ministry of Science and ICT, Korea

Korea

Jan. 2021 - Dec. 2021

Deep Neural Network Based Real-Time Accurate Voice Source Localization using Drones

Ministry of Science and ICT, Korea

Korea

Jun. 2019 - Dec. 2020

Awards, Honors and Scholarships

Jan. 2021 **First Place & Ministerial Award**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

Fall 2020 **Sungkyun Honorable Scholarship (Fall 2020 - Spring 2022)**, Sungkyunkwan University

Korea

Jun. 2019 **Third Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

Fall 2019 **Academic Excellence Scholarship**, Sungkyunkwan University

Korea

Fall 2018 **Academic Excellence Scholarship**, Sungkyunkwan University

Korea

Academic Services

Conference Reviewer

CVPR 2024-2026, ICCV 2025, ECCV 2026, NeurIPS 2024-2025, ICML 2025, ICLR 2025-2026, ACM MM 2024-2025

Journal Reviewer

TPAMI 2026

Patents

“A Method for Inferring of Generating Direction of Sound Using Deep Network and an Apparatus for the Same”

Application No.: 10-2020-0032737

Korea

2020

Skills

Programming

CUDA, Python (PyTorch, TensorFlow), C/C++

Extracurricular Activities

Teaching Assistant

Sungkyunkwan University (SKKU)

Korea

- Operating Systems (Fall 2020)
- Basic data structures and algorithms (Spring-Fall 2019)